Domain 1: Google ADK Native Capabilities & GenAI Orchestration

[ADK-01] Sub-Agent Hand-Off Visibility

When a complex query passes through 6 ADK sub-agents (Supervisor → Orchestrator → Dispatcher → Selector → Planner → Synthesizer), the user must see live step-by-step progress ("Checking Primavera P6 schedule...", "Cross-checking SAP POs...") rather than staring at a frozen spinner for 45 seconds.

[ADK-02] Runaway Agent Loop Protection (max_llm_calls = 45)

When ambiguous project data causes Object_Selector and Query_Planner to loop repeatedly, the runtime must enforce hard token/iteration circuit breakers (before_model_callback) and return a partial grounded synthesis rather than timing out or burning $5 in tokens on a single question.

[ADK-03] User Mid-Flight Cancellation ("Stop Generating")

When an executive realizes they typed "Mumbai DC-1" instead of "Chennai DC-1" and clicks Stop 8 seconds into a 60-second query, the backend must immediately terminate the active ADK Runner task and cancel downstream LLM/SQL calls to prevent wasted compute and session state corruption.

[ADK-04] Cross-Session Executive Memory (MemoryService)

When a BU Head logs in on Wednesday after discussing "Package 4 GIS transformer delays" on Monday, the agent must recall their role, active project focus, and prior action items without forcing them to re-explain context from scratch.

[ADK-05] Versioned Session Artifacts (ArtifactService)

When a Planner runs three "What-If" schedule recovery simulations in a single chat session (v1, v2, v3), the platform must store and retrieve each generated comparison table or chart as a versioned artifact without overwriting earlier iterations.

[ADK-06] Multi-Model Routing & Automatic Failover

When the primary reasoning model (e.g., Gemini 2.5 Pro / Claude / GPT-4o) hits a regional quota spike or latency degradation during a Monday morning review, the ADK model layer must seamlessly fail over to a secondary region or fallback model without dropping the user's chat turn.

[ADK-07] Deterministic Guardrail Callbacks (before_tool_callback)

Before any LLM-generated SQL query executes against Databricks, an ADK before_tool_callback must deterministically inspect the AST to block raw table scans, unauthorized cross-BU joins, or DDL/DML statements.

[ADK-08] Golden Trajectory Regression Testing (adk eval)

Whenever a developer modifies a prompt in build_v42_agents.py or updates a semantic YAML file, CI/CD must automatically verify that the ADK agent still selects the exact right sequence of semantic objects and tools ($\geq$ 95% grounded accuracy, AC-03) before deploying to production.

[ADK-09] Future Cross-BU Agent Federation (A2A Protocol)

When Adani's central Finance or Procurement team deploys their own standalone AI agent next year, Project SLM's ADK Orchestrator must be able to invoke that external agent via standardized Agent-to-Agent (A2A) protocol without rewriting the core platform.

[ADK-10] Hot-Reloading Prompts & Semantic YAMLs

When a domain SME fixes a KPI formula description or business glossary synonym in the Semantic Engine YAML (prism_semantic_layer), the change should take effect within minutes without requiring a full 20-minute container rebuild and cluster redeployment.

Domain 2: Agent Telemetry, Observability & FinOps

(AC-12)

[TEL-01] End-to-End Trace Correlation (UI → ADK → SQL)

When a CFO reports that a ₹120 Cr capex variance number looks wrong, an engineer must be able to paste a single conversation_id and see the exact React request, all 15 ADK sub-agent spans, the exact compiled SQL sent to Databricks, and the SQL Warehouse query profile in one unified trace view within 5 minutes.

[TEL-02] Per-BU & Per-Project Token Chargeback (160M Tokens/Mo)

At group scale across 23 BUs, Finance must receive an automated monthly breakdown of LLM token consumption, Vector Search queries, and Databricks SQL DBU spend attributed accurately to each BU, Project, and User Role.

[TEL-03] Sub-Agent Bottleneck Pinpointing

When complex query latency creeps above the 60s P95 SLA (AC-07), operations teams must immediately see whether the slowdown is happening inside Object_Selector LLM reasoning, Databricks SQL execution, or cold-start queuing.

[TEL-04] Automated Hallucination & Citation Drift Alerts (AC-02, AC-04)

Every production response must be asynchronously scored (or sampled) by an automated LLM-as-a-Judge pipeline to verify that every number in the final synthesis actually appeared in the Evidence_Builder tool output, alerting ops if unsupported claims exceed 5%.

[TEL-05] Sensitive Data Redaction in Telemetry Spans

When an authorized executive queries sensitive L1 commercial bid values or employee safety incident details, those raw values must be masked or access-controlled inside OpenTelemetry / MLflow trace logs so DevOps engineers viewing traces cannot read confidential commercial data.

[TEL-06] Negative Feedback & "Thumbs Down" Replay

When a Project Manager clicks "Thumbs Down" on an answer, the platform must capture the full ADK session state, semantic YAML version, and tool outputs into an evaluation dataset so engineers can replay and fix the exact failure.

[TEL-07] Real-Time Quota & Cost Anomaly Circuit Breaker

If a compromised account or buggy client script triggers 5,000 rapid chat queries from a single user, telemetry must trigger an automatic per-user rate limit before exhausting the BU's monthly token budget.

[TEL-08] Tool Error Rate & SQL Compilation Failure Dashboards

When a source pipeline change causes Query_Planner's generated SQL to fail syntax or schema validation, operations must receive an immediate alert showing which semantic object (e.g., p6_activity_dc) is failing before users complain.

Domain 3: Multi-BU Delta Sharing, Governance & Hiding Raw Data (AC-10, SC-08)

[GOV-01] Blocking Raw Row Dumps While Serving Verifiable Evidence

When a user asks "Give me all rows of the SAP PO line-item table" or tries prompt-injection, the system must refuse to expose raw underlying table dumps while still serving curated, record-level citations (PO #, Vendor Name, Milestone Variance) in the Split-Screen Evidence Panel (04_Split_Chat_Evidence_Panel.png).

[GOV-02] Enforcing Per-User BU & Project Isolation Over Delta Shares (AC-10)

Because Delta Sharing shares a BU's table with the central SLM workspace as a whole, the architecture must enforce per-user BU, Project (Chennai DC-1 vs. Mumbai DC-1), and Function (Commercial vs. Safety) filtering so a user can never query another BU's shared data.

[GOV-03] Preserving Corporate Entra ID Claims Across the Backend → Agent → SQL Hop

Custom corporate Entra ID attributes (BU_Code, Assigned_Projects, DoA_Level, Department) must survive from user login all the way into the ADK Agent and Unity Catalog SQL session without being stripped by platform reverse proxies (a known limitation on Databricks Apps OBO).

[GOV-04] Cross-BU Executive Rollup vs. Single-Site Restriction

When a Group Executive asks "Compare schedule adherence across Data Centre, Thermal, and Airports BUs", the system must seamlessly aggregate across 3 Delta Shares, whereas the exact same prompt from a Site Engineer must automatically scope only to their assigned site.

[GOV-05] Column-Level Masking of Commercial & Margin Data

A Site Planning Engineer and a Chief Commercial Officer must be able to query the exact same Procurement Semantic Object (ariba_po_status), with the Engineer seeing technical delivery dates while unit prices and vendor bid margins are dynamically masked (NULL or REDACTED).

[GOV-06] Graceful Degradation When One BU's Delta Share Is Stale or Broken

If the Thermal BU's Primavera P6 pipeline (DP-10) fails or changes schema overnight, a Group Executive's cross-BU query must still return accurate metrics for Data Centre and Airports while displaying a clear banner: "Thermal BU P6 data is 18 hours stale — excluded from rollup."

[GOV-07] Cross-Cloud / Cross-Region Egress Control on Delta Shares

If BU Data Lakes reside on Azure Databricks (centralindia / southindia) while the ADK Agent runtime sits on GCP (GKE / Vertex AI), the architecture must push heavy joins and aggregations down into Databricks SQL so gigabytes of raw Delta rows never traverse cross-cloud internet links.

[GOV-08] Unstructured Document Entitlement Parity (DP-01 ACC, DP-09 SharePoint)

A user who is blocked from seeing Mumbai DC-1 rows in structured SQL tables must also be strictly blocked from retrieving vector chunks of Mumbai DC-1's contracts, drawings, and NCR PDFs in Vector Search.

[GOV-09] Zero-Copy Governance Without Duplicating 23 BU Warehouses

Adding a new BU via Delta Sharing must apply centralized Unity Catalog Row Filters and Semantic Views over the shared tables without copying terabytes of raw BU data into 23 separate physical databases.

Domain 4: Runtime User File Uploads, Vectorization & Multimodal RAG (Use Cases 3, 14, 16)

[VEC-01] Uploading Large Engineering Documents (>10 MB to 100 MB)

When a Contracts Manager uploads a 35 MB multi-annexure EPC Contract PDF or large BOQ Excel workbook in chat, the platform must not reject the upload with a 413 Payload Too Large error (whereas Databricks Apps imposes a hard 10 MB proxy limit).

[VEC-02] Sub-5-Second Runtime Vectorization (Bypassing Batch Index Lag)

When a user attaches a document and immediately asks a question in the same chat turn, the file must be parsed, chunked, embedded, and searchable in <5 seconds without waiting 5–15 minutes for a batch Databricks DELTA_SYNC vector index pipeline.

[VEC-03] Strict Session Isolation of "What-If" Uploaded Files

When a Planner uploads a speculative "Draft Recovery Schedule v3.xlsx" to simulate milestone shifts (Use Case 3), those vector chunks must be strictly isolated to (user_id, session_id) so the BU Head's concurrent chat never mistakes the draft upload for official project data.

[VEC-04] Cross-Referencing Uploaded Files Against Live Delta-Shared SQL Data

When a user uploads a vendor's delay-claim letter PDF and asks "Are the rain-delay dates claimed in this letter supported by our daily Safex site logs (DP-12) and P6 schedule (DP-10)?", the ADK agent must

jointly query the ephemeral runtime vectors and live Unity Catalog SQL tables in a single turn.

[VEC-05] Complex Table & Engineering Drawing Extraction (OCR / Vision)

Uploaded project files contain multi-page nested BOQ tables, scanned stamp signatures, and drawing revision blocks (DP-01 ACC); runtime parsing must preserve table row/column structure rather than mangling numbers with naive plain-text splitting.

[VEC-06] Controlled Promotion of Verified Uploads to Permanent BU Knowledge Base

When a Project Director reviews a runtime-uploaded Mom/Claim analysis and clicks "Publish to Project Repository", the system must move the document from ephemeral session storage into governed Unity Catalog Volumes and trigger permanent BU-level indexing with full audit metadata.

[VEC-07] Automatic Ephemeral Cleanup & TTL Governance

Abandoned runtime file uploads and temporary session vectors must automatically expire and purge after a configurable policy window (e.g., 24–72 hours) so unverified shadow documents do not accumulate forever.

[VEC-08] CPU/Memory Isolation for Heavy Document Parsing

If 10 users simultaneously upload 40-page PDFs for runtime OCR and chunking, the CPU/memory spike must be isolated in dedicated worker pods or serverless functions so it does not starve the main FastAPI/ADK event loop serving 90 other chat users.

Domain 5: Frontend UX & Dual-Speed Responsiveness (00_SLM_Wireframe.png, AC-06, AC-07)

[UX-01] Dual-Speed Experience (< 2s Dashboard vs. 10s–60s Deep Agent Chat)

As shown in 00_SLM_Wireframe.png, clicking KPI cards (Capex Budget ₹6,240 Cr, Schedule Adherence 68%) or switching BU filters on the dashboard must load in < 2 seconds from an operational cache/Lakebase/Postgres layer, never waiting 30 seconds for an LLM agent or cold SQL Warehouse.

[UX-02] Real-Time Streaming of Intermediate Reasoning & Citations (SSE)

During a 45-second complex root-cause query (AC-07), the Split Chat UI must smoothly stream each ADK sub-agent's intermediate thought badge and progressively populate the right-hand Evidence Panel (04_Split_Chat_Evidence_Panel.png) without proxy buffering.

[UX-03] Zero 502/504 Gateway Timeouts on 60s–180s Deep Queries

Multi-source cross-BU synthesis queries taking 60–180 seconds must never be killed prematurely by ingress load balancers, Azure Front Door, ACA Envoy proxies (240s limit), or Databricks Apps reverse proxies.

[UX-04] Interactive Mind-Map / Knowledge Graph Drill-Down (< 1s Node Expansion)

In the Knowledge Graph view (07_Split_Chat_Knowledge_Graph.png), clicking Delay in Land Acquisition → Impact on Piling → Vendor Claim must expand graph nodes instantaneously from pre-indexed dependency edges while syncing context with the chat pane.

[UX-05] Deep-Linking from Evidence Citations to Original Source Pages

When a user clicks a citation badge ([Drawing Rev-04, Page 12] or [SAP PO #4500192]) in the Evidence Panel, the UI must render a secure inline PDF page preview or source record drawer within 2 seconds using short-lived signed URLs.

[UX-06] Resilient Mobile / Site-Office Reconnection

When a Site Head at a remote construction site (e.g., Khavda or Navi Mumbai site office) experiences a 5-second cellular network blip mid-query, the frontend must automatically reconnect to the active session_id and resume streaming the in-flight ADK answer instead of losing the turn.

[UX-07] Custom Corporate Domain (slm.adani.com) & Seamless SSO

Executives opening https://slm.adani.com on desktop or mobile must land directly in the app via corporate Entra ID SSO without seeing a raw platform URL (*.databricksapps.com), double-login prompts, or platform workspace chrome.

[UX-08] Side-by-Side Scenario Comparison Pinning

When evaluating two recovery options in chat (Option A: Air-freight GIS vs. Option B: Resequence Civil Work), users must be able to pin both agent outputs side-by-side in the UI and export the comparison to PDF/PPTX.

[UX-09] Asynchronous "Notify Me When Ready" for Ultra-Deep Portfolio Audits

If a Group Executive triggers an exhaustive 23-BU portfolio audit or 50-page monthly board pack generation that takes 3+ minutes, they must be able to navigate away to the Dashboard and receive an in-app toast notification when the agent completes.

Domain 6: Backend Orchestration, Controlled Write-Back & Proactive Workflows (AC-09, Use Cases 6, 7, 15)

[BKD-01] Human-in-the-Loop Approval Gate Before Enterprise Write-Back (AC-09)

Under Use Case 15 (Controlled Enterprise Action & Write-Back), when a user asks the agent to update a milestone forecast or create an action item in SAP/Primavera/Clapia, the agent must never execute the write directly—it must stage a draft payload and pause until an authorized human clicks Approve.

[BKD-02] Long-Lived Durable Agent Pause (Hours/Days) Across Approval Routing

If a Project Engineer proposes a schedule baseline change in chat that requires the BU Head's approval under Delegation of Authority (DoA), the paused workflow state must survive in durable storage for hours or days (including across container restarts) and resume once approved.

[BKD-03] 100% Tamper-Proof Before/After Write-Back Audit Trail (AC-09)

Every approved write-back action must atomically record who requested, who approved, timestamp, source agent reasoning, exact old value, and exact new value in an immutable audit table.

[BKD-04] Transactional Idempotency on External ERP/P6 APIs

If a user double-clicks "Approve Write-Back" or an SAP BTP API call times out and retries, idempotency keys must guarantee that duplicate Purchase Requisitions or duplicate NCR records are never created in source systems.

[BKD-05] Event-Driven Proactive Threshold Alerts (Use Case 6)

Without waiting for a user to open chat, when a fresh Delta Share sync reveals that a critical-path milestone slipped by >7 days or a safety NCR was logged in T-Pulse/Safex, a background worker must invoke the ADK agent to draft an impact brief and push a notification to the Project Head.

[BKD-06] Automated Monday 8:00 AM PMO Governance Pack Generation (Use Case 7)

A scheduled cron workflow must run the ADK synthesis pipeline headless across all active projects every Monday morning to pre-generate standardized Weekly Progress Reports (PDF/Word) and executive morning digests.

[BKD-07] Read/Write Workload Separation (Transactional OLTP vs. Analytical OLAP)

High-frequency application writes (saving chat messages, user bookmarks, alert acknowledgments, write-back state machines) must execute in <10ms against an OLTP database (Lakebase / Cloud SQL / Azure Postgres) rather than queuing behind analytical Delta Lake queries.

[BKD-08] Outbound Enterprise Connector Hub (SAP BTP, Primavera P6, Teams/ Email)

The backend runtime must securely reach on-premises / private-cloud SAP BTP, Oracle Primavera P6 APIs, and Microsoft Graph (Teams/Outlook notifications) via private hybrid connectivity.

Domain 7: Scalability & 100-User Burst Performance (AC-11, SC-09)

[SCL-01] The Monday 9:30 AM 100-User Governance Burst (2,500+ LLM Calls)

When 100 project leaders across 23 BUs simultaneously run complex queries during Monday morning review meetings—generating 1,000 to 4,500 LLM calls and 200+ SQL queries within a 60-second window—the platform must keep simple queries $\leq$10s (AC-06) and complex queries $\leq$60s (AC-07).

[SCL-02] Independent Horizontal Autoscaling of Frontend, Backend, and ADK Agent Tiers

Because ADK Agent pods are memory- and I/O-heavy (holding large semantic graphs and 45-step LLM contexts) while Frontend/Backend pods are lightweight, the platform must scale the ADK Agent tier independently to 10–30 replicas without hitting hard platform ceilings (such as Databricks Apps' 4 vCPU / 16 GB RAM and 5-replica cap).

[SCL-03] Zero-Amnesia Pod Scaling & Load Balancing

When horizontal autoscaling spins up 8 new ADK agent pods mid-morning, Turn 2 of an active user's conversation must seamlessly execute on any pod by hydrating state from shared DatabaseSessionService / VertexAiSessionService without losing Turn 1 context.

[SCL-04] Database Connection Pool Protection (PgBouncer / Lakebase Limits)

When 20 autoscaled ADK agent pods each run 20 async coroutines writing session state and telemetry, connection pooling must prevent too many clients already crashes on the underlying PostgreSQL / Lakebase instance.

[SCL-05] Databricks SQL Warehouse Multi-Cluster Scaling & Result Caching

When 100 users ask overlapping portfolio questions, the underlying Databricks Serverless SQL Warehouse must auto-scale clusters instantly and serve identical semantic metric queries from result cache in $<$500ms.

[SCL-06] Provisioned LLM Throughput & Priority Token Queuing

Under peak 100-user load (160M tokens/month), interactive executive chat turns must take priority over background batch tasks (like proactive alert scanning or document indexing) so background jobs never trigger 429 Too Many Requests errors for live users.

[SCL-07] Pre-Warmed Replicas to Eliminate Cold-Start Penalties

At 8:55 AM before business hours begin, the hosting platform must maintain minimum warm instances (min-replicas $>=$ 2) so the first executives logging in do not suffer a 25-second container or model-endpoint cold start.

[SCL-08] Non-Blocking Async I/O Across All Tool Calls

Inside the ADK agent runtime, every LLM call, Databricks SQL execution, and Vector Search lookup must run asynchronously (async/await or thread-pool offloaded) so a single slow 20-second SQL query never blocks 15 other users sharing that container replica.